Part 18 — The pipeline, end to end: twelve problems on the stages the set was missing

You asked whether the 105 problems cover the whole route from choosing a host to shipping a purified product. I audited them against your list instead of answering from memory. Most stages were well covered; three were not covered at all, and two were covered only sideways. This part fills them.

The audit, by pipeline stage

Stage	Problems before	Verdict
Pathway design and yield ceilings	15	Strong
Scale-up and transport	13	Strong
Statistics and design of experiments	13	Strong
Economics and planning	13	Strong
Analytics and data	12	Strong
Enzyme and protein engineering	10	Adequate
Model curation and software	10	Adequate
Extraction and recovery	9	Adequate for extracellular, **nothing on intracellular release**
Fermentation modes	7	Fed-batch and chemostat only; **no mode comparison, no two-stage**
Two-phase and multiphase	7	Adequate
Host and strain selection	6	**No decision problem** — all were single-host calculations
DNA assembly and plasmid construction	3	**Thin** — nothing on copy number, burden or stability
Expression tuning and regulation	2	**Thin** — nothing on combinatorial library sizing
Growth against product optimisation	2	**Thin** — consortia only, no growth coupling
Genome editing (CRISPR, recombineering)	**0**	**Absent entirely**

Zero editing problems in a set meant for someone whose stated interest includes gene editing is the worst of those, and PP5 and PP6 fix it. The twelve here are **PP1 to PP12**, taking the set to **117**.

On "real problems from published papers"

These are built on real cases and I have to be exact about what that means. **I have no network access**, so I cannot open a paper, pull its data or check a number. Where a problem names a published result, the figure is recalled from memory and marked `source: recalled, verify` — treat every one as a claim to check against the paper itself, which is listed. The *arithmetic* in each solution was computed here and is reliable; the *input taken from a paper* is not verified. That distinction is the honest version of what you asked for, and the lookup workbook's sheet 10 is where these belong until you have checked them.

Stage 1 — Choosing the host

PP1 · extends week 59 · 2 h

Lysine: why *Corynebacterium* and not *E. coli*

Case: industrial L-lysine by *C. glutamicum*. Reported around 0.55 g/g at over 120 g/L after systems metabolic engineering (Becker, Wittmann and colleagues). **Source: recalled, verify.**

You must choose a host for a lysine process. The carbon-limited ceiling is 0.75 mol/mol; allowing the pentose phosphate pathway to supply the four NADPH per lysine gives 0.81 mol/mol.

1. Convert both ceilings to g/g, and express the reported industrial yield as a fraction of each.
2. *E. coli* has no dedicated lysine exporter and *C. glutamicum* has LysE. Compute the broth titer a retained product could reach at 50 and 100 gDCW/L if the intracellular concentration caps at 200 mM.
3. List the three properties that actually decide this choice, and name what you give up.
4. Under what change of product would the answer flip to *E. coli*?

Worked solution. 0.75 mol/mol is **0.609 g/g** and 0.81 is **0.657 g/g**, so a reported 0.55 g/g is **84%** of the cofactor-corrected ceiling — close enough that further yield work has little room, which is itself the finding.

The export arithmetic is decisive. With 2 mL of cell volume per gDCW, a product held at 200 mM intracellular gives only **2.9 g/L at 50 gDCW/L and 5.8 g/L at 100** — two orders of magnitude below the 120 g/L the process needs. **Without an exporter the process is impossible at any yield**, which is why this is a host-selection problem and not a pathway problem.

The three deciding properties: a native exporter; tolerance to 120 g/L of a charged amino acid in the broth; and the absence of overflow metabolism, since *C. glutamicum* does not make acetate at high glucose the way *E. coli* does. What you give up is cycle time and toolkit depth — *E. coli*'s design–build–test loop is faster, which matters for the twenty iterations a new pathway needs.

The answer flips for a **secreted protein, a non-natural small molecule needing many heterologous enzymes, or anything where the campaign needs dozens of iterations** — there the toolkit outweighs the exporter, and if the product is uncharged and diffuses out, the export advantage disappears entirely.

What it teaches: host choice is usually settled by one property that the pathway cannot fix — here, export — and the yield comparison is a distraction until that is checked.

PP2 · extends weeks 18, 19 and 40 · 2.5 h

1,4-butanediol: designing a pathway that nature does not have

Case: 1,4-BDO in *E. coli* via succinyl-CoA and 4-hydroxybutyrate. Reported 18 g/L in the original paper and over 100 g/L commercially. **Source: recalled, verify.**

BDO is $C_4H_{10}O_2$ (90.12). The route runs glucose → oxaloacetate → succinate → succinyl-CoA → succinate semialdehyde → 4-hydroxybutyrate → 4-HB-CoA → BDO.

1. Compute the carbon and mass yield ceiling with and without anaplerotic CO_2 fixation.
2. Count the reductant the route consumes per BDO and compare with what glycolysis supplies.
3. The route needs an enzyme activity that did not exist. Which step, and how would you find one?
4. 18 g/L is what fraction of the ceiling, and what would you measure to decide whether to push titer or yield?

Worked solution. Four carbons out of six gives **0.500 g/g** with two CO_2 lost; with PEP carboxylase re-fixing CO_2 the balance improves to about **0.667 g/g** — and which applies depends on how much carbon enters through the carboxylation rather than pyruvate dehydrogenase, so the ceiling is a *design variable*, not a constant.

Reductant is the binding constraint. The reductive steps from succinyl-CoA to BDO consume roughly **4 NAD(P)H per molecule** while glycolysis to oxaloacetate supplies **2** — a deficit of two, which must come from further oxidation of carbon (lowering yield), from formate oxidation, or from a different split between the oxidative and reductive branches. That is the quantitative reason this product is hard, and it is visible before any strain exists.

The missing activity is the **CoA-dependent reduction of 4-HB to the aldehyde**, and the route to one is the chapter S3 and part 8 chapter 29 route: find enzymes acting on an analogous substrate (aldehyde dehydrogenases on acyl-CoAs), screen homologues, then evolve. 18 g/L against a 0.500–0.667 g/g ceiling and a typical 100 g/L target is **roughly**

a fifth of a commercially viable titer, so the next measurement is a carbon balance with by-products: if carbon is going to acetate and ethanol, the problem is the fermentative balance, not the pathway.

What it teaches: a novel pathway's hardest number is its redox deficit, and it is computable from the stoichiometry before a single gene is cloned.

Stage 2 — Building it: plasmid, expression, editing

PP3 · extends weeks 10 and 32 · 1.5 h

Plasmid or chromosome, decided by arithmetic

A seed train from a 1 mL vial to a 100 m³ vessel is about 27 generations. Three plasmid classes are available, with illustrative loss rates and burdens, against single-copy integration.

1. Compute the generations in the seed train, then plasmid retention at 27 and 40 generations for each class.
2. A 300-copy plasmid gives how much more gene dose than one integrated copy? What would it take to match it from the chromosome?
3. Which option would you choose for a 48-hour fed-batch, and which for a 30-day continuous run?
4. What single measurement decides it in practice?

Replicon	Loss/division	Burden	Retention at 27 gen	at 40 gen
pUC-type, high copy	10^{-3}	15%	**75%**	**32%**
pBR322-type, medium	5×10^{-4}	8%	95%	88%
pSC101-type, low	2×10^{-4}	3%	99%	98%
Chromosomal, single copy	—	—	100%	100%

Worked solution. $\log_2(10^8)$ = **26.6 generations** from vial to vessel. The high-copy plasmid is already a quarter lost before production begins and two-thirds lost by 40 generations — and the cells that lost it grow 15% faster, so they dominate the end of the run, which is where most of your product is made.

On dose: one integrated copy against 300 is a 300-fold deficit, and no promoter closes that — the strongest promoters span perhaps 10- to 100-fold over a medium one. Realistically you combine **a stronger promoter (10×) with three integrated copies (3×)** for 30 copy-equivalents, and accept that a plasmid remains ahead on dose while losing on stability.

So: for a **48-hour fed-batch** a medium-copy plasmid with selection is defensible — 95% retention over the train, and the run is short. For a **30-day continuous run** only integration survives the arithmetic of problem Z8. The deciding measurement is cheap and almost never done: **plate the end of a pilot run with and without antibiotic and count both**. The ratio is your retention, and if it is below about 90% the yield you measured is not the yield of your strain.

What it teaches: plasmid choice is a race between gene dose and retention over the generations the process actually runs, and both sides are measurable in an afternoon.

PP4 · extends weeks 8 and 28 · 2 h

Balancing three enzymes: how big must the library be?

Case: combinatorial promoter and ribosome-binding-site libraries for pathway balancing, as used for mevalonate and carotenoid pathways. **Source: recalled, verify.**

1. Compute the number of combinations for 3 enzymes × 5 levels, 3 × 8, 4 × 5 and 5 × 4.
2. For each, how many clones give 95% coverage?
3. What coverage do 96 and 384 wells give for the 125- and 625-member libraries?
4. If expression effects are lognormal with σ = 0.6, how much better than the median is the best of 24, 96 and 384 random clones?
5. Design the campaign you would actually run.

Design	Combinations	Clones for 95% coverage
3 enzymes × 5 levels	125	373
3 × 8	512	1,532
4 × 5	625	1,871
5 × 4	1,024	3,066

Worked solution. Coverage is the problem X20 calculation: 96 wells of a 125-member library covers **54%**, 384 covers **95%**, and 384 of a 625-member library covers only **46%**. So a single 96-well plate does not survey even the smallest

useful library — the common practice of picking one plate and reporting the best clone is sampling half the design space.

But the returns are blunt: the best of 24 random clones is **3.4-fold** above the median, the best of 96 is **4.6-fold**, and the best of 384 is **6.0-fold**. Sixteen times the screening effort buys 1.8 times the improvement, because you are sampling the tail of a lognormal.

The campaign that follows: **three enzymes at five levels, 384 clones, in two rounds** — one plate to estimate the spread and check the assay, then three more if the spread justifies it. Beyond three enzymes, move to a designed subset rather than full factorial coverage, and use the measured strengths as model inputs (chapter 41's surrogate) instead of screening the product space.

What it teaches: library size is set by coverage arithmetic, and the gain from screening more is logarithmic — so the design, not the plate count, is where the improvement comes from.

PP5 · extends weeks 23 and 24 · 2.5 h

CRISPR as a counterselection: what fraction of your colonies are edited?

Case: Cas9-assisted recombineering in bacteria, and multiplex automated genome engineering (MAGE), which reported 4.3 billion variants in three days. **Source: recalled, verify.**

In bacteria Cas9 is used to kill the unedited rather than to cut and repair, because there is no efficient end-joining. Recombination supplies the edit; Cas9 removes the background.

1. For recombination frequencies of 10^{-4}, 10^{-3} and 10^{-2}, and killing of 99%, 99.9% and 99.99%, compute the edited fraction among survivors and the clones to screen for a 95% chance of one.
2. If 1 in 10^4 cells escapes by losing the target site without being edited, what fraction of survivors are false positives at a recombination frequency of 10^{-3}?
3. For MAGE, compute the fraction of cells edited at all of 4, 10 and 24 loci after 20 cycles at per-cycle efficiencies of 10%, 20% and 30%.
4. When would you choose each method?

Recombination	Kill 99%	Kill 99.9%	Kill 99.99%
10^{-4}	1.0% (299 clones)	9.1% (32)	50% (5)
10^{-3}	9.1% (32)	50% (5)	90.9% (2)
10^{-2}	50% (5)	91% (2)	99% (1)

Worked solution. The edited fraction is f/(f + (1−f)(1−kill)), and the table shows why counterselection is transformative: at a recombination frequency of 10^{-4} — typical for a point mutation by oligo recombineering without selection — you would screen 30,000 colonies, and three nines of killing cuts that to **32**. The practical lesson is that **killing stringency matters as much as recombination efficiency**, and it is the easier of the two to improve.

Escapers set the floor. At 10^{-4} escape and 10^{-3} recombination, **9.1% of survivors are false positives** — colonies that lost the protospacer or the guide rather than taking the edit — so verification by sequencing is not optional, and a construct whose guide can be silently deleted will give you exactly this.

MAGE after 20 cycles reaches **95.5% of cells edited at all 4 loci at 20% per-cycle efficiency, 89.1% at 10 loci, and 75.7% at 24**; at 10% efficiency those fall to 27.4% at ten loci and 4.5% at twenty-four. So MAGE is for **many loci, tolerated imperfectly, with a selection or screen at the end**, while Cas9-assisted recombineering is for **one precise edit you must have in a clean background**.

What it teaches: editing is a probability calculation, and the number of colonies you must screen follows from recombination frequency, killing stringency and escape rate — all three measurable.

PP6 · extends weeks 23 and 32 · 2 h

Integrating a cassette, verifying it, and removing the marker

Case: λ-Red recombineering with FLP-removable markers, and *sacB* counterselection. **Source: recalled, verify.**

1. With 1 µg of a linear cassette, a transformation efficiency of 10^7 cfu/µg and recombination at 10^{-4} or 10^{-5}, how many integrant colonies do you expect?
2. You then plate 10^8 cells to counterselect marker loss on sucrose. At *sacB* escape rates of 10^{-3}, 10^{-4} and 10^{-6}, and 100 or 1,000 true excisants, what fraction of survivors are real?
3. Design the verification PCRs. Why is one PCR insufficient?

4. What does FLP removal leave behind, and what does that cost on the second round?

Worked solution. Expected integrants are **1,000 colonies at 10^{-4} and 100 at 10^{-5}** — ample, which is why λ-Red works at all.

The counterselection is where it goes wrong. Plating 10^8 cells with a *sacB* escape rate of 10^{-3} gives 10^5 escapers against 100 true excisants: **0.10% of survivors are real, so you screen about 1,000 colonies for one**. At 10^{-4} escape it is 101 colonies; only at 10^{-6} does it become **2 colonies per hit**. Two fixes follow directly: use a counterselection whose escape rate you have measured, and **plate a smaller fraction of the culture** — re-plating 1% cuts escapers a hundredfold while costing you proportionally fewer true events, which is favourable whenever escapers dominate.

Verification needs **two junction PCRs**, each with one primer in the chromosome outside the homology arm and one inside the cassette, plus one internal amplicon. One PCR cannot distinguish a correct single insertion from a tandem duplication or from a cassette that integrated elsewhere by a secondary homology. FLP removal leaves a **34 bp FRT scar**; a second round at a different locus leaves a second scar, and the two scars are substrates for recombination between them — which is the mechanism by which a strain with several rounds of marker removal can delete the intervening chromosome.

What it teaches: the integration step is easy and the marker-removal step is where the screening burden hides; the ratio of escapers to true events is the number to compute before plating.

Stage 3 — Making it produce

PP7 · extends week 39 · 2 h

Growth coupling for succinate: the deletions and what they cost

Case: anaerobic succinate strains of *E. coli*, built by deleting the competing fermentative routes and then evolving. **Source: recalled, verify.**

Using the 40-reaction core model anaerobically on 10 mmol of glucose, with growth-coupled production defined as the *minimum* product flux at maximum growth.

1. Compute the wild-type guaranteed succinate flux.
2. Evaluate ΔPYK, ΔPYK ΔACK, ΔPYK ΔACK ΔPFL, and ΔADH ΔLDH.
3. Express each as mol per mol of glucose, and state the growth cost.
4. Which design would you build, and what would the published strains have added?

Design	Growth (h^{-1})	Guaranteed succinate	mol/mol glucose
Wild type	0.217	0.00	0.00
ΔPYK	0.134	8.23	0.82
ΔPYK ΔACK	0.072	9.05	0.90
ΔPYK ΔACK ΔPFL	0.048	9.36	0.94
ΔADH ΔLDH	**lethal anaerobically**		

Worked solution. The wild type guarantees nothing: it can grow without making any succinate, so selection does not favour producers. Each deletion closes more routes and forces more product, at a growth cost that compounds — **0.82 mol/mol at 62% of wild-type growth, rising to 0.94 mol/mol at 22%**. ΔADH ΔLDH is lethal because it removes the cell's only remaining NADH sinks, which is the kind of answer a model gives cheaply and a strain-building campaign discovers expensively.

I would build **ΔPYK ΔACK**: 90% of the maximum coupled yield at nearly twice the growth rate of the triple, which matters because a slow strain means a longer fermentation and a bigger seed train. What the published strains added beyond the model: **evolution**, to recover growth rate after the deletions, and work on the **carboxylation step and glucose uptake** — neither of which this model can suggest, since its only uptake route is the PEP-consuming phosphotransferase system.

What it teaches: growth coupling is a trade curve, not a design, and the right point on it is chosen by the fermentation economics rather than by the maximum.

PP8 · extends weeks 44 and 56 · 2 h

Two-stage fermentation: where to put the substrate

A fixed budget of 200 g/L of glucose and 48 hours. Growth yield 0.45 g/g at $\mu = 0.45\ h^{-1}$; in stage two the cells do not grow and convert substrate to product at 0.40 g/g with a specific rate of 0.25 $g \cdot gDW^{-1} \cdot h^{-1}$. Start from 0.5 gDW/L.

1. Write the time needed for each stage as a function of the fraction f of substrate spent on growth.
2. Tabulate biomass, product and total time for f from 0.05 to 0.7.
3. Find the f that maximises product within 48 hours.
4. Compare with a one-stage growth-coupled process at 0.30 g/g.

f to growth	Biomass (gDW/L)	Growth time	Product (g/L)	Production time	Total
0.05	4.5	4.9 h	76.0	67.6 h	**72.4 h — too slow**
0.10	9.0	6.4	72.0	32.0	38.4 h
0.20	18.0	8.0	64.0	14.2	22.2 h
0.50	45.0	10.0	40.0	3.6	13.6 h

Worked solution. Substrate spent on cells is substrate not in the product, so product falls linearly with f — but the *time* to convert the rest falls as 1/f, because more catalyst works faster. The optimum sits where the 48-hour limit binds: **f = 0.078, giving 7.0 gDW/L, a switch at 5.9 h, and 73.8 g/L of product in exactly 48 hours** — an overall yield of

0.369 g/g and a productivity of **1.54 g/L/h**.

Against a one-stage growth-coupled process at 0.30 g/g, which gives 60 g/L and 1.25 g/L/h, the two-stage route is **23% better in titer and productivity** — because it spends almost no carbon on cells. The caveats are the ones the arithmetic does not contain: non-growing cells lose productivity over time, which this model ignores, and the switch has to be triggered by something real (nitrogen exhaustion, a temperature shift, an inducer), each of which has its own lag.

What it teaches: the two-stage switch time is a constrained optimum between carbon spent on catalyst and the time that catalyst needs, and it lands where the clock binds.

PP9 · extends week 44 · 2.5 h

Batch, fed-batch or continuous, for the same strain

One strain: μ_{max} 0.45 h^{-1}, K_s 0.5 g/L, $Y_{X/S}$ 0.45, specific productivity 0.30 $g \cdot gDW^{-1} \cdot h^{-1}$ falling linearly to zero at 60 g/L of product, maintenance 0.02. Turnaround between batches is 12 hours.

1. Simulate a batch from 200 g/L of glucose: titer, time, productivity, with and without turnaround.
2. Simulate an exponential fed-batch at $\mu = 0.12$ h^{-1} from a 500 g/L feed.
3. Compute the continuous steady state at $D = 0.05$ to 0.3 h^{-1} on a 50 g/L feed.
4. Tabulate all three on titer, yield, productivity and vessel-time, and choose.

Mode	Titer	Time	Productivity	With turnaround
Batch	46.4 g/L	14.3 h	3.26 g/L/h	1.77 g/L/h
Fed-batch, $\mu = 0.12$	53.7 g/L	34.7 h	1.55 g/L/h	1.15 g/L/h
Continuous, $D = 0.1$	60.0 g/L	—	**6.00 g/L/h**	6.00 g/L/h
Continuous, $D = 0.2$	33.5 g/L	—	6.70 g/L/h	6.70 g/L/h

Worked solution. Continuous wins on productivity by a factor of three to four, because it never stops and never spends time at low cell density — and that is exactly why it is rare. Its titer is capped at the steady-state product concentration, so the downstream train sees a dilute stream; it needs 30 to 60 days of genetic stability, which problem Z8 says a cheater at 8% advantage destroys; and contamination over that period is a serious risk.

Fed-batch reaches the **highest cell density (82 gDW/L)** and the highest titer of the two closed modes, at the cost of productivity, because most of the run is spent accumulating biomass. Batch is the most productive closed mode here and the simplest, but it leaves no room to control overflow.

The choice: **fed-batch** for a product that is inhibitory and needs titer for the separation — which is most of them — and **continuous** only when the organism is stable, the product is not inhibitory, and the downstream can take a dilute stream. The decisive comparison is not productivity but **cost per kilogram delivered**, which pulls toward titer.

What it teaches: mode selection is a four-way trade between titer, productivity, stability and downstream cost, and the mode with the best productivity is usually the wrong answer.

Stage 4 — Getting it out

PP10 · extends chapter S1 and week 56 · 2 h

An intracellular product: how many homogeniser passes?

Case: PHA recovery by solvent extraction, hypochlorite digestion, enzymatic digestion or surfactant. **Source: recalled, verify — the recovery and purity figures below are illustrative.**

Disruption releases product first-order in passes: released fraction = $1 - e^{-kn}$. The product also loses activity or quality at a fixed rate per pass.

1. Tabulate release after 1, 2, 3 and 5 passes for k = 0.6, 0.9 and 1.4.
2. With k = 0.9 and 3% or 8% quality loss per pass, compute the net recovered yield.
3. Find the optimum number of passes at 8% loss.
4. For a PHA at 60% of cell mass and 100 gDW/L, compare the four recovery routes on delivered product.

Passes	Release at k = 0.9	Net at 3% loss/pass	Net at 8% loss/pass
1	59%	58%	55%
2	83%	79%	71%
3	93%	85%	**73%**
5	99%	**85%**	65%

Worked solution. Release saturates while damage accumulates, so the net has an interior maximum: at 8% loss per pass the optimum is **3 passes for 73%**, and a fifth pass actively destroys product. At 3% loss the optimum is flat from three to five passes. Running "until lysis is complete" is therefore wrong whenever the product is fragile — and the measurement that settles it is a pass-by-pass assay of *active* product, not of released protein.

For the PHA, 60% content at 100 gDW/L is **60 g/L of polymer in the cells**; at 90% recovery you deliver 54 g/L. The four routes differ less in recovery (85–95%) than in what they cost elsewhere: solvent extraction gives the best purity and the worst solvent handling; hypochlorite is cheap and **degrades the polymer's molecular weight**, which for a material product is the specification, not a detail; enzymatic digestion preserves it and costs enzyme; surfactant is cheapest and leaves 8% impurity. The choice is made by the product specification, not the recovery column.

What it teaches: for an intracellular product the release step has an optimum, and for a polymer the recovery method changes the product itself rather than just how much you get.

PP11 · extends weeks 45 and 49 · 2 h

An extracellular product: two-phase capture through to crystals

Case: amorphadiene and artemisinic acid production with an organic overlay, then recovery. Partition coefficient around 107. **Source: recalled, verify.**

1. For solvent fractions of 0.02, 0.05, 0.10 and 0.20, compute the broth concentration relative to a single phase and the fraction of product held in the solvent.
2. Choose a solvent fraction and justify it against the displaced working volume.
3. With phase separation 99%, solvent stripping 97%, crystallisation 85% and drying 99%, compute the train recovery.
4. Give the overall recovery from fermenter to dry product, and name the step to improve first.

Solvent fraction	Broth concentration, relative	Product in the solvent
0.02	0.321	69%
0.05	0.159	85%
0.10	0.086	92%
0.20	0.045	96%

Worked solution. Even 2% solvent pulls **69%** of the product out of the aqueous phase and cuts what the cells see to a third, which is the whole point of in situ removal — relief of product inhibition comes cheaply. Going from 5% to 20% solvent moves the captured fraction from 85% to 96% while displacing four times the working volume, so **5% is the**

sensible choice: the last 11% of capture costs 15% of your vessel.

The downstream train multiplies to **80.8%**, and with 95% captured into the solvent the overall recovery from fermenter to dry product is **76.8%**. The step to improve first is **crystallisation at 85%**, which is both the worst step and the one with the most headroom — a second crop from the mother liquor typically recovers half of what is lost, taking the train to about 87% for the cost of one more vessel and a filtration.

What it teaches: in situ removal is bought cheaply at low solvent fraction, and the recovery train's worst step is where the next engineering hour belongs.

PP12 · extends weeks 44 and 49 · 2.5 h

A whole process at scale: 1,3-propanediol at 135 g/L

Case: bio-based 1,3-propanediol, reported around 135 g/L at about 0.51 g/g from glucose in an engineered *E. coli*. **Source: recalled, verify.**

1. Compute broth volume and water to remove per kg of product, and the feedstock cost at $0.40/kg glucose.
2. At 30 gDW/L and 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$, compute OUR, the required k_La and the heat duty in a 100 m^3 vessel.
3. Propanediol boils at 214 °C and water at 100 °C. Compute the energy and cost of removing the water.
4. Compare separation with feedstock cost, and compute what the water cost would have been at 50 g/L.

Worked solution. At 135 g/L the broth is **7.4 L per kg** of product with **6.4 kg of water to remove** — against 20 kg at 50 g/L, which is the entire reason the titer target was set where it was. Feedstock is 1/0.51 = **1.96 kg of glucose per kg**, so **$0.78/kg** at $0.40/kg sugar.

Oxygen: OUR = **120 mmol·L^{-1}·h^{-1}**, needing a k_La of **750 h^{-1}** — at the top of what a large vessel achieves, so this process is close to its transfer limit and oxygen enrichment or a lower feed rate is likely in the real plant. Heat is **1.53 MW**, needing 132 m^3/h of cooling water and **307 m^2 of heat-transfer area** against roughly 110 m^2 of jacket: internal coils, as problem X12 predicted.

Water removal is **14.5 MJ per kg** of product, or **$0.12/kg** at $8/GJ — **13% of the variable cost** alongside $0.78 of feedstock. At 50 g/L it would have been **$0.34/kg**, nearly tripling. So the headline achievement of this process is not the yield, which is close to the theoretical maximum anyway, but the **titer**, and the number it bought is a separation cost.

What it teaches: a real process is judged on four numbers at once — yield for feedstock, titer for separation, rate for capital, and oxygen for the vessel — and titer is usually the one that pays for itself twice.

Closing notes to part 18

Where the set now stands against your pipeline

Stage	Before	Added	Now
Host and strain selection	6	PP1	7, with a decision problem
Pathway design	15	PP2	16, including a novel-enzyme route
Plasmid and construction	3	PP3	4, with copy number against stability
Expression tuning	2	PP4	3, with library sizing
Genome editing	**0**	PP5, PP6	**2**
Growth against product	2	PP7	3, with growth coupling
Fermentation modes	7	PP8, PP9	9, with two-stage and a mode comparison
Recovery, intracellular	**0**	PP10	**1**
Recovery, extracellular and purification	9	PP11	10, as a full train
Scale-up	13	PP12	14, with a whole process costed
Total problems	**105**	**12**	**117**

What is still not covered, honestly

- **Protein products** — secretion capacity, inclusion bodies, refolding, glycosylation. The set is built around small molecules and polymers. If your target labs do therapeutic proteins, this is the next gap to fill.
- **Regulatory and quality** — GMP, validation, batch records. Named in part 12 and never worked as a problem, because it is procedure rather than calculation.
- **Anything needing a real instrument or a real strain** — covered as far as synthetic data allows (R1 to R5), and no further.
- **Verified published numbers.** Every paper figure in this part is recalled, not checked. The arithmetic built on them is sound; the inputs need an afternoon with the papers and sheet 10 of the lookup workbook.

The eight results worth remembering

- **PP1** — without an exporter, a retained product caps at 2.9 g/L at 50 gDCW/L. Host choice decided by one property.
- **PP3** — a high-copy plasmid is 25% lost before production begins and 68% lost by 40 generations.
- **PP4** — 96 wells covers 54% of a 125-member library, and the best of 384 clones is only 1.8 times better than the best of 24.
- **PP5** — three nines of Cas9 killing cuts the screen from 30,000 colonies to 32.
- **PP6** — *sacB* at 10^{-3} escape means 1,000 colonies screened per true excisant.
- **PP7** — ΔPYK ΔACK gives 0.90 mol/mol at 0.072 h^{-1}; the triple buys 4% more product for half the growth.
- **PP9** — continuous is 3 to 4 times more productive and almost always the wrong choice.
- **PP12** — at 135 g/L water removal costs $0.12/kg; at 50 g/L it would cost $0.34/kg.